

MASTER DOCUMENTS TREE

Ultra-Safe Central Capsule

Complete Engineering Documentation Structure • Approximately 115 Documents
Baseline derived from Document 00 (Technical Specification) + Full System Requirements

This Master Documents Tree defines the complete set of engineering documents required to design, analyse, manufacture, integrate, test, certify and support the Ultra-Safe Central Capsule (~7 tonne operational mass, carbon-fibre 12 cm roll-cage, 4-layer parachute & separation architecture, mechanical landing legs, life-support and multi-platform integration). Documents are organised by engineering domain. Document numbers follow the pattern USC-XXX-NNN.

1. SYSTEM REQUIREMENTS & CONCEPT DEFINITION (12 documents)

- USC-SYS-001 System Requirements Specification (SRS) – Top Level
- USC-SYS-002 Mission & Operational Concept Document
- USC-SYS-003 Interface Control Document (ICD) – Host Platforms (Ground / Aerial / Maritime)
- USC-SYS-004 Functional Architecture Description
- USC-SYS-005 Safety Requirements Specification (SRS-Safety)
- USC-SYS-006 Environmental Requirements (Temperature, Vibration, Shock, Salt Fog, etc.)
- USC-SYS-007 Human Factors & Occupant Accommodation Requirements
- USC-SYS-008 Reliability, Availability, Maintainability (RAM) Requirements
- USC-SYS-009 Concept of Operations (CONOPS) – Normal, Emergency & Recovery
- USC-SYS-010 Trade Study Report – Capsule Size & Mass Variants (Compact / Standard / Heavy)
- USC-SYS-011 System Design Description (SDD) – Overall
- USC-SYS-012 Configuration Management Plan

2. PRIMARY STRUCTURE & ROLL-CAGE (14 documents)

- USC-STR-001 Structural Design Criteria & Load Cases (Impact, Crash, Flight, Ground)
- USC-STR-002 Carbon-Fibre Roll-Cage Design Description (12 cm primary tubes, 6-axis)
- USC-STR-003 Roll-Cage Detailed Design Drawings & 3D Model Package
- USC-STR-004 Node Joint Design (Titanium Grade 5 + Structural Adhesive)
- USC-STR-005 Finite Element Analysis (FEA) Report – Static Strength
- USC-STR-006 FEA Report – Dynamic Crash / Impact (12-15 g)
- USC-STR-007 FEA Report – Modal & Vibration Analysis
- USC-STR-008 Outer Shell & Multi-Layer Skin Design Description
- USC-STR-009 Glazing Design (Polycarbonate / Multi-layer Impact Resistant)
- USC-STR-010 Materials Allowables & Qualification Data Package (CFRP, Ti, Elastomers)
- USC-STR-011 Manufacturing Process Specification – CFRP Roll-Cage
- USC-STR-012 Structural Test Plan (Static, Dynamic, Drop)
- USC-STR-013 Structural Test Results & Correlation Report
- USC-STR-014 Damage Tolerance & Repair Manual

3. LANDING GEAR, IMPACT ABSORPTION & ENERGY MANAGEMENT (11 documents)

- USC-LND-001 Landing System Requirements & Performance Specification
- USC-LND-002 Mechanical Leg Design Description (6 fixed legs)
- USC-LND-003 Hydraulic / Pneumatic Shock Absorber Design Specification
- USC-LND-004 Elastomer Pad Design & Material Specification (15-20 cm thick)
- USC-LND-005 Inter-Capsule to Host Shock-Mount Interface Design
- USC-LND-006 Energy Absorption Analysis Report (Multi-stage)
- USC-LND-007 Landing Dynamics Simulation Report
- USC-LND-008 Drop-Test Plan & Instrumentation Specification
- USC-LND-009 Drop-Test Results & Model Correlation

USC-LND-010 Landing Leg Separation Mechanism (Pyrotechnic / Mechanical)

USC-LND-011 Maintenance & Overhaul Manual - Landing System

4. PARACHUTE & RECOVERY SYSTEM (12 documents)

USC-PAR-001 Parachute System Requirements Specification
USC-PAR-002 Canopy Design Description (6 Primary + 8 Auxiliary)
USC-PAR-003 Deployment Sequence & Logic Description (4-Layer Architecture)
USC-PAR-004 Layer-1 Electrical Deployment System Design
USC-PAR-005 Layer-2 Independent Pyrotechnic Path Design
USC-PAR-006 Layer-3 Pure Mechanical (Inertia / Manual) Path Design
USC-PAR-007 Layer-4 High-Pressure Nitrogen Ejection System Design
USC-PAR-008 Pilot / Extractor Chute Design & Integration
USC-PAR-009 Parachute Packing, Rigging & Installation Manual
USC-PAR-010 Deployment Simulation & Trajectory Analysis
USC-PAR-011 Parachute System Test Plan (Ground, Aerial, Failure Modes)
USC-PAR-012 Parachute System Qualification Test Report

5. HOST SEPARATION & PYROTECHNIC SYSTEMS (9 documents)

USC-SEP-001 Separation System Requirements & Safety Criteria
USC-SEP-002 Separation Mechanism Design Description (Pyro + Hydraulic + Mechanical)
USC-SEP-003 Umbilical & Frangible Interface Design (Electrical / Hydraulic / Data)
USC-SEP-004 Pyrotechnic Device Specification & Safety Data
USC-SEP-005 Separation Sequence Timing & Interlock Logic
USC-SEP-006 Separation Dynamics & Clearance Analysis
USC-SEP-007 Separation System Test Plan (Ground & In-Flight Simulation)
USC-SEP-008 Separation System Qualification Report
USC-SEP-009 Safe Handling, Storage & Disposal Procedure - Pyrotechnics

6. LIFE SUPPORT, ENVIRONMENTAL CONTROL & SURVIVABILITY (10 documents)

USC-LSS-001 Life Support System Requirements (72-hour autonomy)
USC-LSS-002 Oxygen Storage & Delivery System Design (with CO₂ scrubbing)
USC-LSS-003 Emergency Water & Food Storage Specification
USC-LSS-004 Air Filtration & Toxic Atmosphere Protection Design
USC-LSS-005 Thermal Control / Passive Phase-Change System Design
USC-LSS-006 Cabin Atmosphere Control & Monitoring System
USC-LSS-007 Emergency Lighting & Visual Acquisition System
USC-LSS-008 First-Aid & Medical Kit Specification
USC-LSS-009 Life Support System Test & Endurance Report
USC-LSS-010 Occupant Survival Manual (Post-Landing Procedures)

7. FIRE DETECTION & SUPPRESSION (6 documents)

USC-FIR-001 Fire Detection & Suppression Requirements
USC-FIR-002 Dual-Agent System Design (Clean-Agent Interior + Foam/Powder Exterior)
USC-FIR-003 Multi-Sensor Detection Logic (Thermal / Smoke / IR)
USC-FIR-004 Nozzle Layout & Coverage Analysis
USC-FIR-005 Fire System Test Plan & Results
USC-FIR-006 Post-Fire Inspection & Recharge Procedure

8. SEATS, HARNESS & OCCUPANT PROTECTION (7 documents)

USC-SEA-001 Seat System Requirements (Energy Absorption & Comfort)
USC-SEA-002 Viscoelastic Seat Design Description (45-50 cm stack)
USC-SEA-003 Five-Point Harness Design & Automatic Lock / Rapid Release
USC-SEA-004 Occupant Injury Criteria Analysis (HIC, Nij, etc.)
USC-SEA-005 Seat Dynamic Test Plan (Sled / Drop)
USC-SEA-006 Seat Dynamic Test Results & Correlation
USC-SEA-007 Seat Installation, Adjustment & Maintenance Manual

9. EMERGENCY EXITS & EGRESS SYSTEMS (6 documents)

USC-DOR-001 Emergency Exit Requirements (Top / Bottom / Left / Right)
USC-DOR-002 Door Mechanism Design (Pyrotechnic / Hydraulic Actuation)
USC-DOR-003 Safety Interlock Logic (G-load & Fire Status Permission)
USC-DOR-004 Egress Path Analysis & Human Factors Assessment
USC-DOR-005 Door System Test Plan & Results
USC-DOR-006 Emergency Exit Operation & Training Manual

10. ELECTRICAL, POWER, AVIONICS & BACKUP SYSTEMS (11 documents)

USC-ELE-001 Electrical System Architecture & Power Budget
USC-ELE-002 Emergency Battery System Design (Li-Titanate / equivalent, ≥ 96 h)
USC-ELE-003 Power Distribution & Isolation Design
USC-ELE-004 Sensor Suite Specification (Accelerometers, Gyros, Fire, Separation)
USC-ELE-005 Flight / Survival Computer Hardware & Software Requirements
USC-ELE-006 Software Design Description & Safety-Critical Software Plan
USC-ELE-007 Backup Path Logic & Fail-Operational Design
USC-ELE-008 Electromagnetic Compatibility (EMC) Control Plan
USC-ELE-009 Wiring Diagrams & Harness Design Package
USC-ELE-010 Electrical System Test & Validation Report
USC-ELE-011 Battery Safety, Charging & Thermal Management Manual

11. COMMUNICATIONS, LOCATION & RECOVERY AIDS (6 documents)

USC-COM-001 Dual Satellite Communication System Design (Iridium + Emergency Starlink)
USC-COM-002 EPIRB & AIS Integration Specification
USC-COM-003 Antenna Deployment & Coverage Analysis
USC-COM-004 Visual Acquisition System (Strobe / High-Intensity LED)
USC-COM-005 Recovery Hard-Point Design (Crane / Helicopter Hook)
USC-COM-006 Communications System Test & End-to-End Verification Report

12. HOST PLATFORM INTEGRATION (8 documents)

USC-INT-001 Integration Requirements – Ground Vehicles
USC-INT-002 Integration Requirements – Helicopters / Rotorcraft
USC-INT-003 Integration Requirements – Fixed-Wing Aircraft / Jets
USC-INT-004 Integration Requirements – Aerial Cranes & Heavy Platforms
USC-INT-005 Integration Requirements – Maritime / Sea Pyramid Platforms
USC-INT-006 Mechanical / Structural Interface Drawings (All Host Types)
USC-INT-007 Electrical & Data Interface Control Documents (per Host Class)
USC-INT-008 Integration Test & Compatibility Verification Plan

13. SYSTEM ANALYSIS, SIMULATION & MODELLING (8 documents)

USC-ANL-001 System Performance Simulation Plan
USC-ANL-002 Multi-Body Dynamics Model (Capsule + Legs + Parachutes)
USC-ANL-003 Crashworthiness & Occupant Safety Simulation Report
USC-ANL-004 Thermal Analysis Report (Normal & Emergency)
USC-ANL-005 Reliability Prediction & FMEA / FMECA Report
USC-ANL-006 Fault Tree Analysis (FTA) – Critical Safety Functions
USC-ANL-007 Monte-Carlo Deployment Success Probability Analysis
USC-ANL-008 Model Validation & Verification Report

14. SYSTEM TEST, VERIFICATION & VALIDATION (9 documents)

USC-TST-001 Master Test & Evaluation Plan
USC-TST-002 Component Qualification Test Matrix
USC-TST-003 Subsystem Integration Test Procedures
USC-TST-004 Full-Capsule Ground Test Plan (Static, Dynamic, Environmental)
USC-TST-005 Drop-Test & Crash-Test Campaign Plan
USC-TST-006 Parachute Aerial Deployment Test Plan
USC-TST-007 Environmental Qualification Test Report
USC-TST-008 System Verification Matrix (Requirements Traceability)
USC-TST-009 Final System Validation & Acceptance Test Report

15. SAFETY, CERTIFICATION & AIRWORTHINESS (7 documents)

USC-SAF-001 System Safety Program Plan
USC-SAF-002 Hazard Analysis & Risk Assessment Report
USC-SAF-003 Safety Case / Safety Assessment Report
USC-SAF-004 Certification Basis & Means of Compliance (FAA / EASA / Military)
USC-SAF-005 Continuing Airworthiness Instructions
USC-SAF-006 Occupant Safety Demonstration Report
USC-SAF-007 Independent Safety Review Board Report

16. MANUFACTURING, QUALITY & SUPPLY CHAIN (7 documents)

USC-MFG-001 Manufacturing Plan & Build Sequence
USC-MFG-002 Quality Management Plan (AS9100 / equivalent)
USC-MFG-003 Special Process Specifications (CFRP Lay-up, Bonding, Pyro Assembly)
USC-MFG-004 Supplier Control & Incoming Inspection Plan
USC-MFG-005 Production Tooling & Jig Design Package
USC-MFG-006 First Article Inspection (FAI) Report
USC-MFG-007 Production Cost Model & Make-or-Buy Analysis

17. INTELLECTUAL PROPERTY & PATENT DOCUMENTATION (6 documents)

USC-IP-001 Invention Disclosure Package – 4-Layer Parachute Architecture
USC-IP-002 Invention Disclosure Package – 6-Axis CFRP Roll-Cage
USC-IP-003 Invention Disclosure Package – Multi-Path Separation System
USC-IP-004 Invention Disclosure Package – Multi-Stage Energy Absorption
USC-IP-005 Freedom-to-Operate (FTO) Search Report
USC-IP-006 Patent Family Filing Strategy & Priority Document Set

18. PROGRAM MANAGEMENT, LOGISTICS & SUPPORT (6 documents)

USC-PM-001 Program Management Plan & Work Breakdown Structure (WBS)
USC-PM-002 Risk Management Plan & Risk Register
USC-PM-003 Integrated Master Schedule (IMS)
USC-PM-004 Logistics Support Analysis & Spares Provisioning
USC-PM-005 Technical Publications Plan (Manuals Suite)
USC-PM-006 Training Needs Analysis & Training Material Package

Category	Documents
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TOTAL ENGINEERING DOCUMENTS	115

This Master Documents Tree constitutes the complete engineering documentation framework required to take the Ultra-Safe Central Capsule from Document 00 baseline through detailed design, analysis, prototype manufacture, qualification testing, certification and production readiness. Document numbers and titles may be refined as the program matures; the overall structure and count of approximately 115 documents is considered sufficient for a full engineering program of this complexity.

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